TerraTrace

Planetary Analog OS

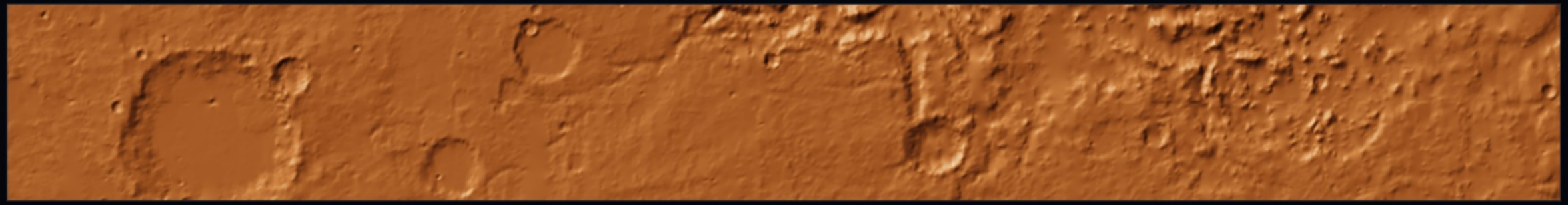
S — SUPPORTED
V — NOT SUPPORTED
R — SUPPORTED

Terrain morphology compared across Earth, Mars, the Moon, Mercury and Ceres at a declared common spatial support. Pareto dominance — never a single score.

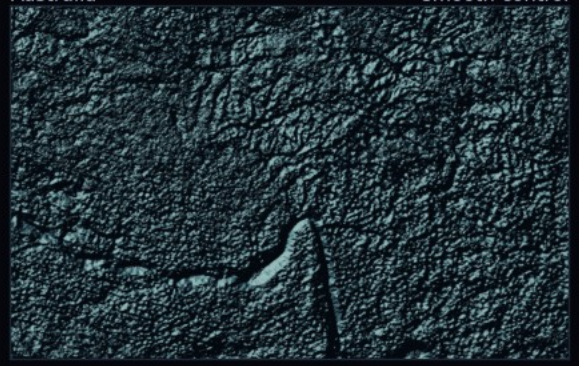
protocol v3, 15/15 confirmatory cells
pre-specified, not pre-registered

Mars MOLA 463 m/px reference terrain

reference terrain



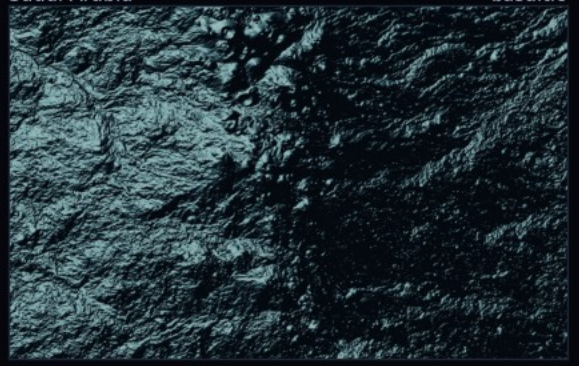
Nullarbor Plain σ 5 m
Australia smooth control



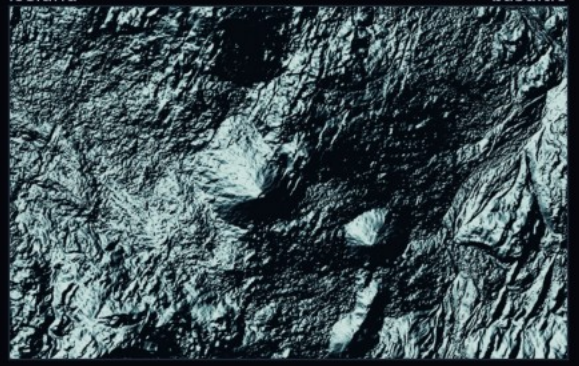
Deccan Traps σ 32 m
India basaltic



Harrat Khaybar σ 102 m
Saudi Arabia basaltic



Odadahraun σ 182 m
Iceland basaltic



Earth candidates · Copernicus GLO-30, 30 m posting

Shading: one light direction and one vertical scale across every panel — nothing is contrast-stretched, so flat terrain reads as flat.

roughness tracks the class; volcanic origin does not

891 offline tests · 51 scientific-correctness oracles · 15 cited sources · MIT
github.com/pal-123456789/terratrace

Morphology only. No traversability, site-access, geologic-equivalence or mission-readiness claim.

NASA SPACE APPS CHALLENGE 2026

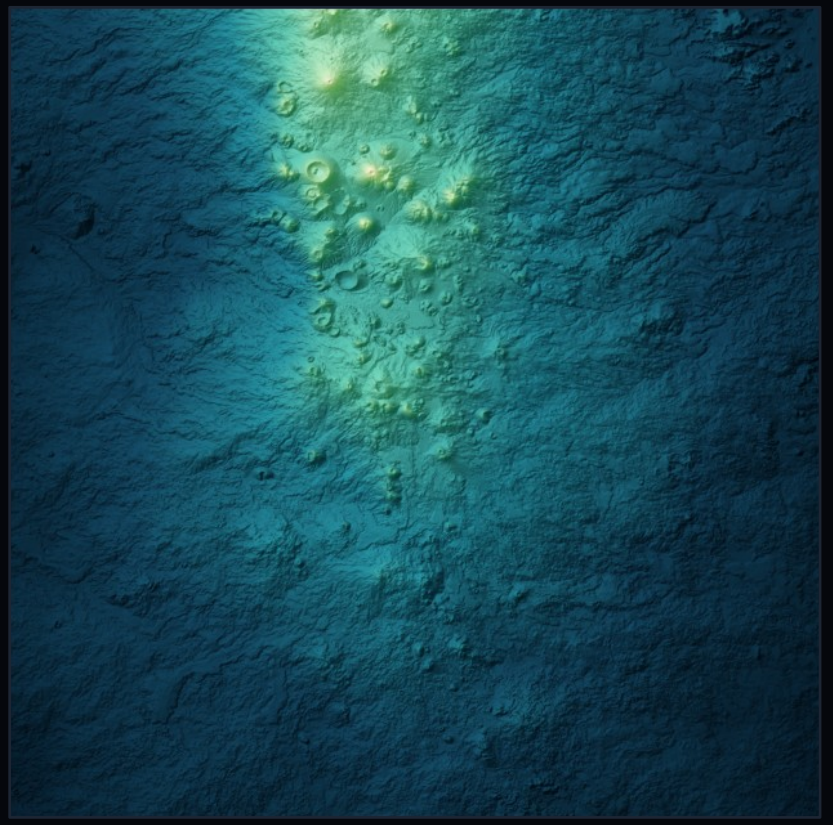
Earth terrain that answers to Mars and the Moon

TerraTrace asks one narrow, measurable question: **which Earth terrains are morphologically comparable to a given planetary reference, at a declared common spatial support?** It answers with measured distance, uncertainty and a gap ledger — and nothing else.

Challenge

“Identify Earth Locations that Analog the Permanent Moon Base Locations and Mars”

Published as open prior art, deliberately, weeks ahead of the event — stated openly rather than dressed up as event work.



Harrat Khaybar, Saudi Arabia · σ 102 m · real Copernicus GLO-30 terrain

PIPELINE

Real elevation to a defensible comparison

Five stages, each one a place where an analog study usually overclaims.

01

Catalog

15 products with coverage, licence and vertical accuracy — body, CRS and datum asserted by tests.

02

Support

Each window reprojected onto its own body's metric grid. Nominal posting (30 m GLO-30, 463 m MOLA) is NOT the compared resolution: the common effective support is 3x source spacing, declared

03

Describe

40 cited descriptor families — ~290 scalar values — computed in pure NumPy.

04

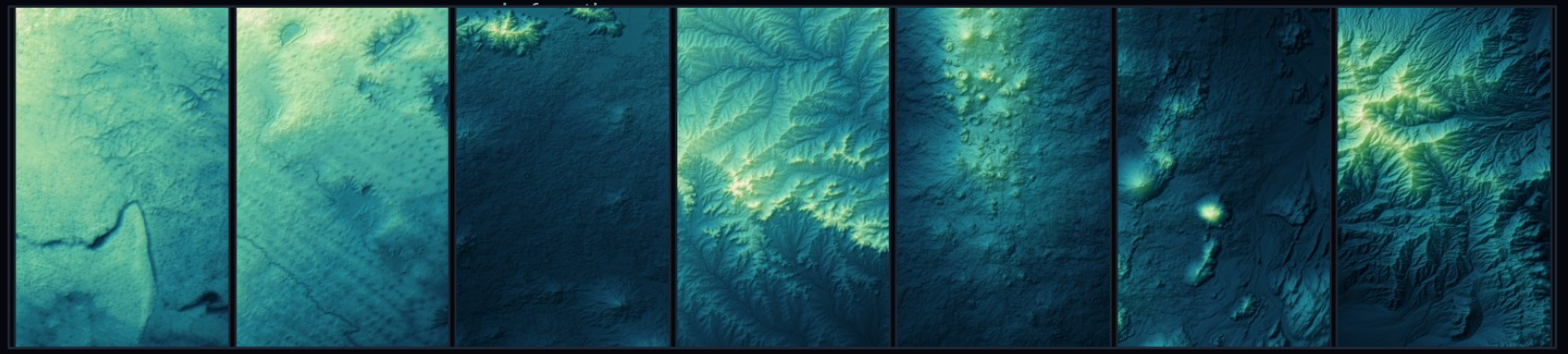
Compare

Pareto dominance across axes chosen per lens. A single composite score does not exist in the code.

05

Refuse

Eligibility filters run first; a gap ledger and a provenance graph travel with every result.



Input is real map data, not a synthetic stand-in — shown here at 30 m posting.

- Earth
- Mars
- Moon
- Mercury
- Ceres

Five bodies, one method.

THE MEASUREMENT LAYER

Depth where it is cited, restraint where it is not

Every descriptor carries a definition, a unit, a window and a citation — and a stated claim boundary.

40

descriptor families, each cited to the literature it comes from

~290

scalar outputs emitted by the descriptor bank

6

decision lenses — five by default, hydro opt-in

0

composite scores anywhere in the codebase

Axes, not a verdict

- Slope, relief, ruggedness, connectivity, curvature, vector ruggedness, roughness/Hurst, openness, geomorphons, power spectrum, anisotropy.
- The decision is Pareto dominance over a small declared set per lens — so a candidate that wins on roughness cannot be quietly traded against one that wins on relief.

Five result states, three of them refusals

- RECOMMEND_FOLLOW_UP and TIE, plus INDETERMINATE (insufficient evidence) and NOT_COMPARABLE.
- BLOCKED is a real outcome: a water or flat tile is filtered out before it can be scored.
- Eligibility filtering runs BEFORE comparison: a blocked unit is never scored, and a study that cannot separate candidates is allowed to say so.

THE BENCHMARK, INCLUDING THE PART THAT WENT AGAINST US

We withdrew our own headline finding

Three acts, recorded in the order they happened: Act II is pre-registered, Act III is pre-specified.

Act I · exploratory

A real multi-region atlas suggested an analog we liked.

Published 2026-10-11. Screened as a hypothesis — never as a result.

Act II · pre-registered

A matched-footprint test to falsify it.

Protocol frozen 2026-10-16 before any v2 result. It failed; the claim was withdrawn.

Act III · pre-specified

Protocol v3 asked what actually explained the match.

Rules and config predate the 2026-11-05 run. The protocol document does not.

Protocol v3, confirmatory cells

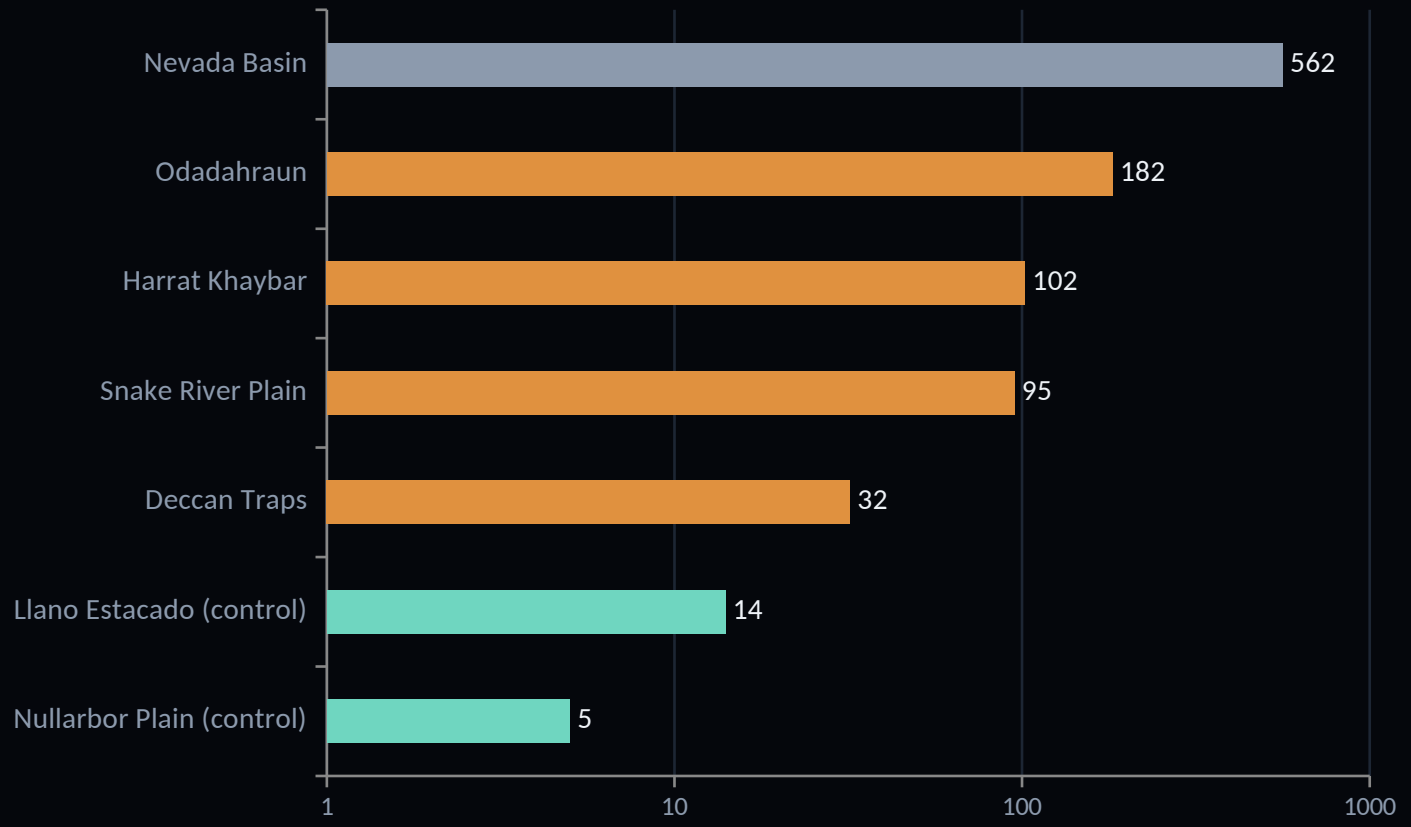
15/15 honest cells · 0 comparability-gate violations · 11 unchanged · 1 destabilised · 3 membership-changed · 0 reversed

All six classes, none hidden: 11 unchanged, 1 destabilised, 3 membership-changed, and 0 stabilised, 0 reversed, 0 error.
Pre-specified, not fully pre-registered: the rules and config predate the run; the protocol document does not.

15 / 15

THE RESULT

Within this panel and lens: roughness tracks the class, origin does not



σ = measured roughness of each Earth candidate at 60 km support (teal = smooth non-volcanic control, amber = basaltic, grey = neither; log scale). S = smoothness/shortlist, V = volcanic specificity, R = resolution robustness.

S — SUPPORTED

A smooth non-volcanic control holds Pareto layer 1 at every honest support, for all three volcanic-plains references.

V — NOT SUPPORTED

It did not establish the pre-declared volcanic-specific advantage, over this panel and lens.

R — SUPPORTED

Of 15 confirmatory cells: 11 unchanged, 1 destabilised, 3 membership-changed, 0 reversed. The destabilised cell is mars_solis at 1500 m.

Where it is soft

The top pick is a near-tie; it reverses on a 6-px crop shift. The class result is robust, the ranking is not.

EVIDENCE

Claims are cheap; these are checked

Every number below is produced by a command in the repository, and the documentation guard fails the build if the prose disagrees with the code.

891

offline tests — no network, no fixtures downloaded, run by a dependency-free runner

51

scientific-correctness oracles over analytic surfaces with known answers

2

independent runs that reproduce the canonical artifact hash

What the gates actually do

- A comparability gate refuses to display any value below the scale that supports it.
- A claim-firewall test fails the build if a forbidden claim appears in the output.
- A documentation-consistency guard recomputes every published count from source and names the file when prose drifts.

Continuous integration on three Pythons

- 3.10, 3.11 and 3.12 with NumPy only — the suite, an optional-dependency hygiene check that runs every test mentioning an optional package, the 51 oracles, and the doc guard.
- An end-to-end job builds the study and the viewer and uploads the bundle. These are repeatable software outputs — they are NOT independent scientific replication.

`e8b823e = the tested implementation commit; the deck is rebuilt after it · CI: github.com/pal-123456789/terratrace/actions`

A keyless path to real 30 m Earth data

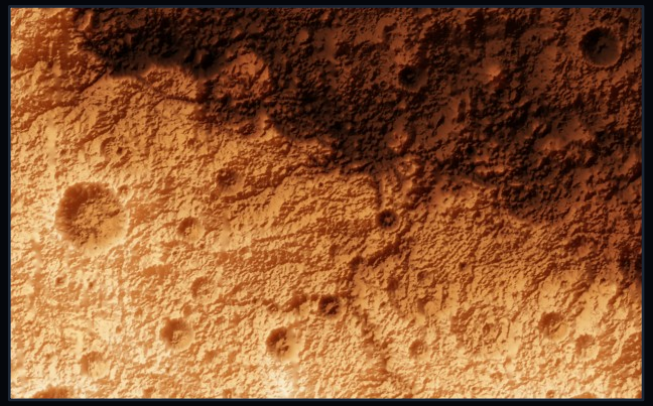
The benchmark runs with no credentials at all.

Copernicus GLO-30 on AWS Open Data

Proven bit-identical to the OpenTopography COP30 product — maximum absolute difference 0.00 m, 100 % of pixels identical. The API key is now optional, not required.

```
One command, no credential.
```

- 15 catalogued products, each with coverage geometry, licence and terms, vertical accuracy and nodata policy — and a per-file SHA-256 in a source lock.
- An evidence graph records every source, transform and unit behind every visible number.
- The whole canonical flow runs offline; no live dependency sits in the demo path.



Mars reference terrain · MOLA, 463 m/px nominal

Independent toolchain check

3x3 relief: within 0.0001 m
Slope means: within 0.016 deg
Reprojection: vs closed-form geodesy, within 0.00003 m
Cell-size and window-anchor assumptions: documented differences
External reviewer: none yet

Five bodies, each reprojected onto its own metric grid and compared at a common effective support, never at nominal posting.

The firewall is the feature

Never claimed, by design

- Rover traversability, autonomy or mission readiness.
- Geological equivalence between two bodies.
- Site access, permitting, ecology or community impact.
- Any single “Mars-likeness” score.

This is an analog-screening instrument. It narrows a search; it does not certify a site.

Open, and named rather than hidden

- Two descriptors re-derived independently; no external reviewer yet.
- No demo video; the script for it ships in the repository.
- All six lunar sites screened and the answer survives the DEM error ensemble — but it is exploratory, and the roughest sites have no close match.
- No external user walkthrough yet; the boundary is argued, not tested.

Every open item is listed first in the traceability page, not buried.

Why this matters for the challenge

A shortlist is only useful if the person reading it can see exactly how far the evidence goes. A tool that will not say “this is a safe place to land” is more trustworthy for saying so than one that quietly implies it.

WHERE TO FIND IT

Everything is public, offline, and reproducible

Reproduce the headline result

```
pip install numpy rasterio
python -m terratrace.cli study --config config/matched_footprint_v3.yml --
source cog
```

Offline once the tiles are cached: the viewer, the reports and the study package need no network. Fetching the DEMs for the first time does.

Read it

- The site, the v3 results page, the generated st viewer.
- The paper, validation report and limitations.
- The traceability page against the project's own master plan.
- The exploratory lunar demonstration on real LOLA terrain.



Repository

The honest ask

The engine is prior art, built weeks ahead of the event. What a team adds from here is the part that counts: a second terrain family it has never seen, an independent re-derivation of one window by a different GIS toolchain, or a validation set drawn by someone with planetary field experience. If you would rather find the place where this quietly overclaims than add a feature, that is the most useful thing you could do to it.